

CSMFAB000000000033 V2.0

Aegis Embark: Thorax-Calm Smart Seatbelt Cover

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log

- STF updated: SiO₂ nanoparticle suspension, yield stress 50 kPa confirmed (2025 data)
- Response time: < 0.5 ms (vs 2 ms V1.0) — critical for crash event coverage
- Crash pressure distribution: 3.3× improvement over naked webbing
- FBG optical respiration and heart rate monitoring added
- All metallic seatbelt hardware replaced with PEEK CF40 equivalent

1. Two Threats, One Cover

Kinetic threat: High-g crash events (>30g, duration 100 ms) compress the thorax against standard 50 mm nylon webbing, which delivers peak contact pressure of ~250 kPa over the clavicle — causing rib fractures in 28% of belted crash occupants (NHTSA 2024).

Electromagnetic threat: Standard seatbelt hardware (metal buckle, metal pre-tensioner housing) forms a conductive loop across the torso. During a GIC event, this loop carries induced current directly against the body.

2. Shear-Thickening Fluid Technology

Composition: 60 vol% fumed SiO₂ (200 nm particles) in polyethylene glycol (PEG-400)

Rheology: - Zero-shear viscosity (at rest): $\eta_0 = 0.8 \text{ Pa}\cdot\text{s}$ — flexible, conforming - Critical shear rate: $\dot{\gamma}_c = 200 \text{ s}^{-1}$ - Post-critical viscosity: $\eta_{\text{peak}} = 85 \text{ Pa}\cdot\text{s}$ at $\dot{\gamma} = 10,000 \text{ s}^{-1}$ - Yield stress: $\tau_y = 50 \text{ kPa}$ (2025 measurement, V2.0)

At crash impact (typical $\dot{\gamma} \approx 50,000 \text{ s}^{-1}$): STF solidifies in < 0.5 ms, creating a rigid-body contact pad that distributes the webbing load over 3.3× greater surface area.

Crash pressure distribution: - Naked webbing: $\sigma_{\text{contact}} = 250 \text{ kPa}$ over $30 \text{ cm}^2 \approx 750 \text{ N}$ concentrated - STF cover activated: $\sigma_{\text{contact}} = 75 \text{ kPa}$ over $100 \text{ cm}^2 \approx$ same 750 N distributed $\approx$ sub-fracture threshold

3. Optical Bio-Feedback (V2.0)

Two FBG sensors sewn into cover fabric: - Respiratory rate: ± 0.5 breath/min accuracy from thoracic expansion detection - Heart rate: ± 2 BPM from aortic pulse pressure wave (detects microstrain in webbing) - Data via PMMA POF to driver display - Zero metallic sensing elements — GIC-safe

4. Non-Metallic Hardware Replacement

All standard metallic seatbelt hardware replaced: - Buckle tongue: PEEK CF40 (tensile strength 220 MPa, crash load 15 kN — passes ECE R17) - Pre-tensioner housing: BFRP

composite (no metal trigger mechanism — pyrotechnic retained but ceramic-isolated from frame) - Height adjuster: Zirconia ceramic slide (no metal rail) - D-ring: Alumina ceramic, 50 mm bore, rated to 20 kN

End CSMFAB000000000033 V2.0 | Carrington Storm Motors